

Know Before It Shows:

Fatigue-aware BCI to improve user engagement

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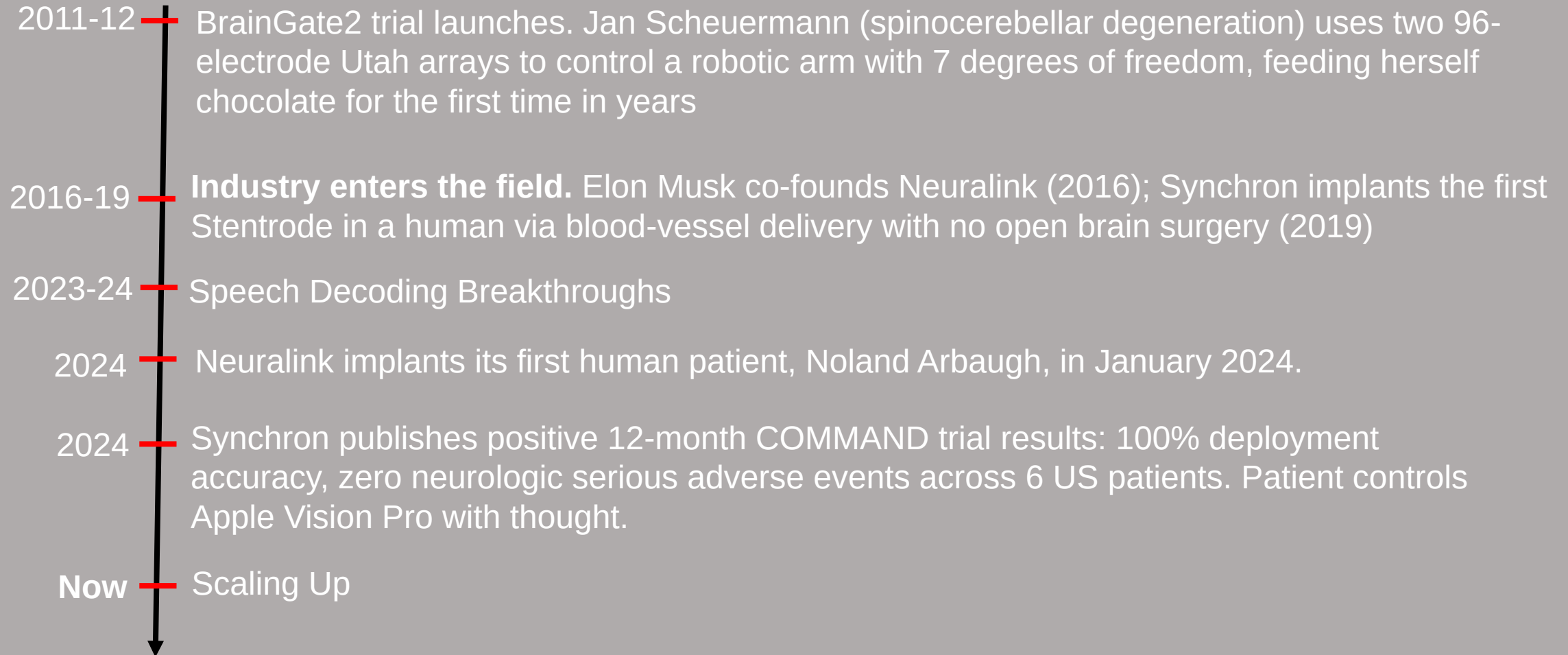
Today we will cover ...

- Brain-Computer Interfacing
 - Background
 - Neuroergonomics
 - A general pipeline
 - Some Examples
- Need for mental state monitoring → Mental Fatigue
-
- Our Approach

History

- 
- 1924 — First human EEG recorded by Hans Berger in Jena, Germany.
 - 1957 — John Eccles wins the Nobel Prize for work on synaptic transmission, establishing the ionic basis of single-neuron signaling.
 - 1969 — Eberhard Fetz (U. Washington) demonstrates operant conditioning of single-neuron firing rates in monkeys, showing that individual neurons can be volitionally controlled. (Fetz, E.E. (1969))
 - 1973 — Jacques Vidal (UCLA), first coins the term “brain-computer interfaces”. (Vidal, J.J. (1973))
 - 1976/77 — First working demo from UCLA
 - 1988 — Farwell and Donchin demonstrate the P300 speller – First practical demonstration
 - 1998 — Philip Kennedy and Roy Bakay implant the first long-term intracortical BCI electrode in a locked-in ALS patient (Johnny Ray), who learns to move a cursor with thought
 - 2004 — First BrainGate Trials – Matthew Nagle controls computer cursor and prosthetic hand via BCI
 - 2006 — BrainGate outcome: Robotic arm control published in Nature. (Hochberg, L.R., et al. (2006))

History



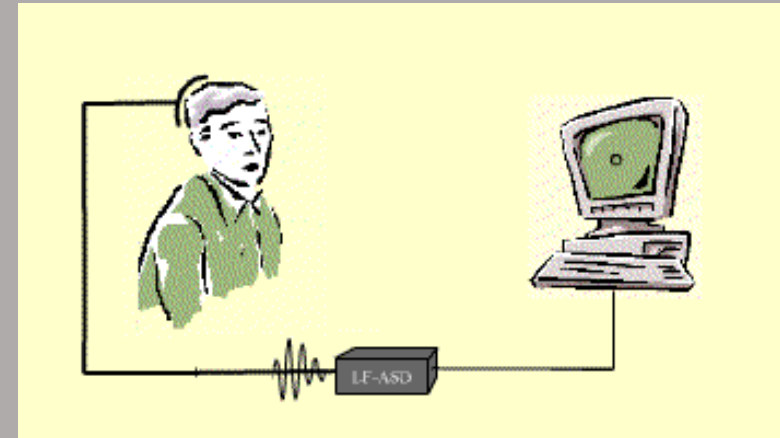
Definitions

Vidal's (1973)

"Can observable electrical brain signals be put to work as carriers of information in man-computer communication for the purpose of controlling such external apparatus as prosthetic devices?"

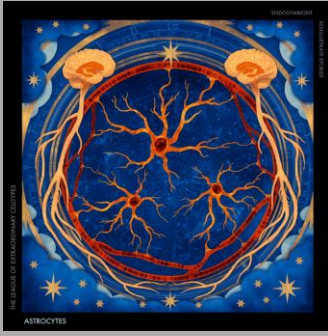
Wolpaw et al. (2002)

A BCI is a communication system in which messages or commands that a person sends to the external world do not pass through the brain's normal output pathways of peripheral nerves and muscles.

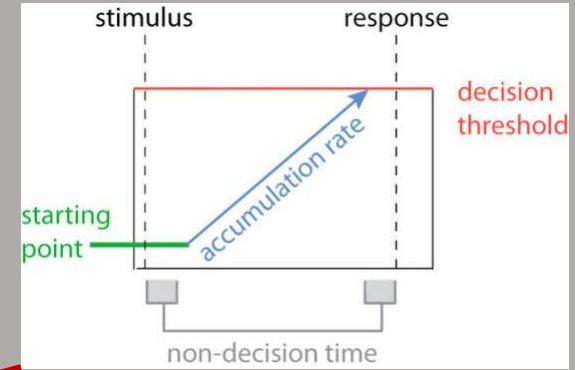
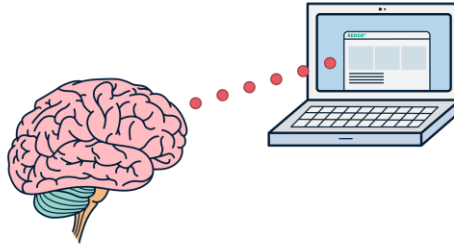


<http://www.ece.ubc.ca/~garyb/BCI.htm>

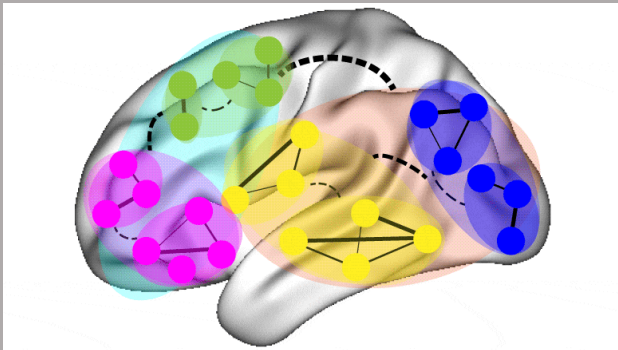
The Connection



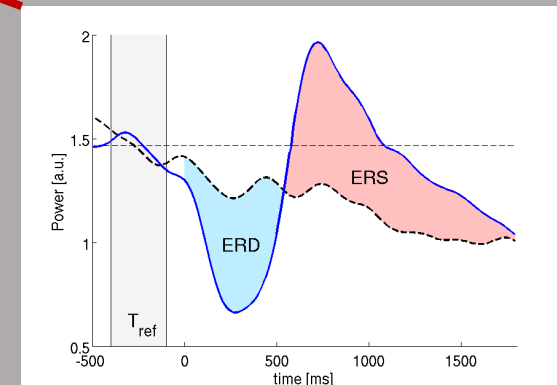
**Neurovascular
Signal Layer**



**Evidence
Accumulation**

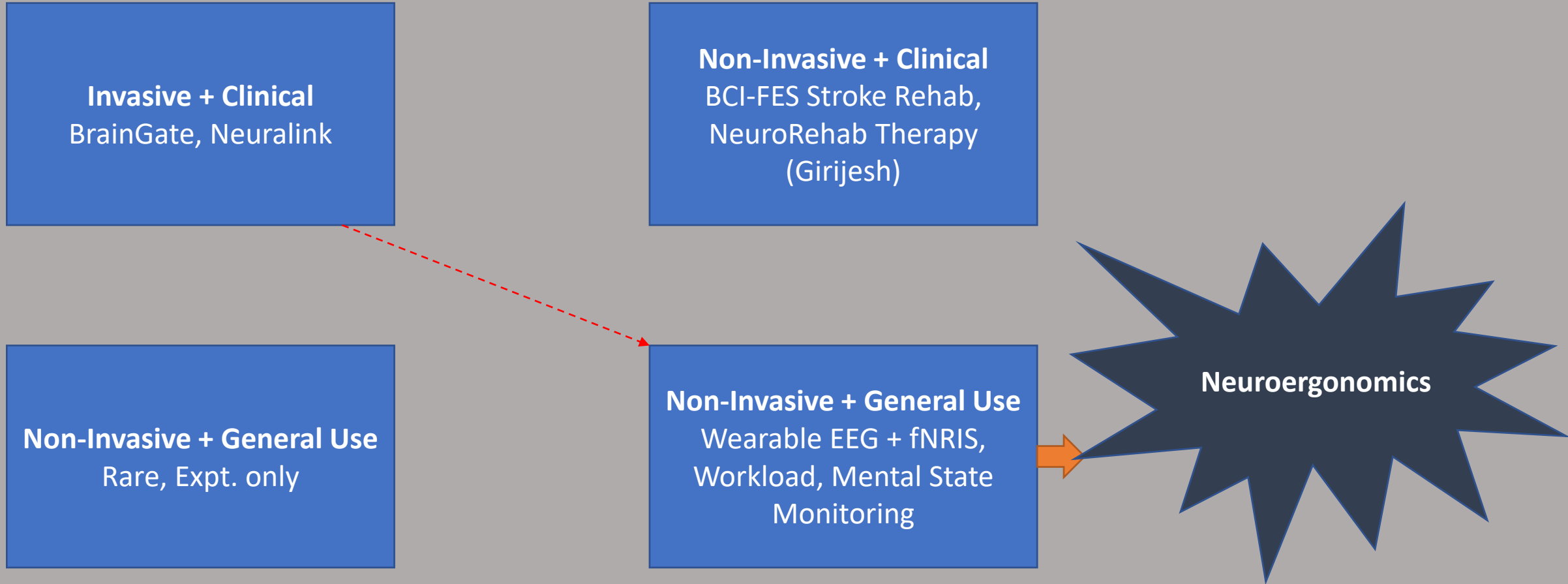


**Brain
Connectivity**



Neuroimaging

Where BCI is headed



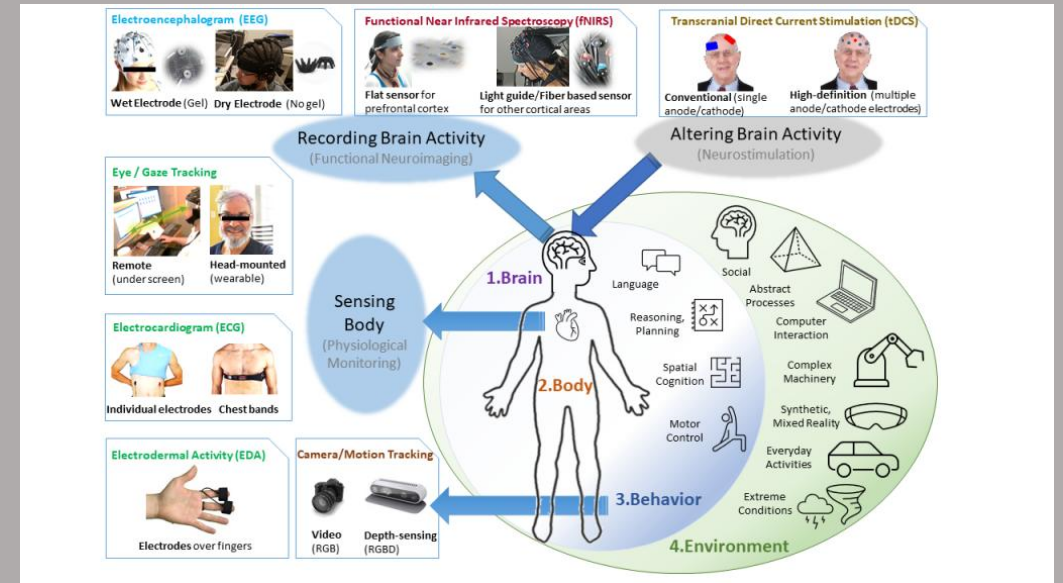
Neuroergonomics

BCI came from clinical necessity, neuroergonomics came from human-factors/ergonomics research trying self-reporting.

Parasuraman (2003)

the study of brain and behavior at work

- The understanding of the brain functioning and its utilization for real world applications is the next frontier.
- DARPA AugCog (2001)
https://en.wikipedia.org/wiki/Augmented_cognition - same non-invasive sensors used clinically, redeployed for real-time operator cognitive-state monitoring

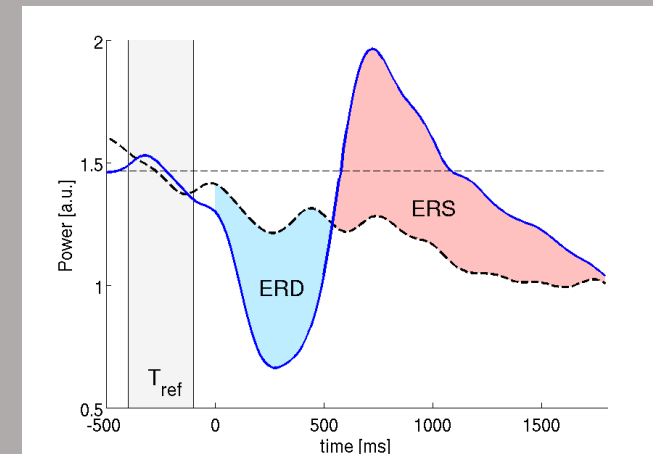
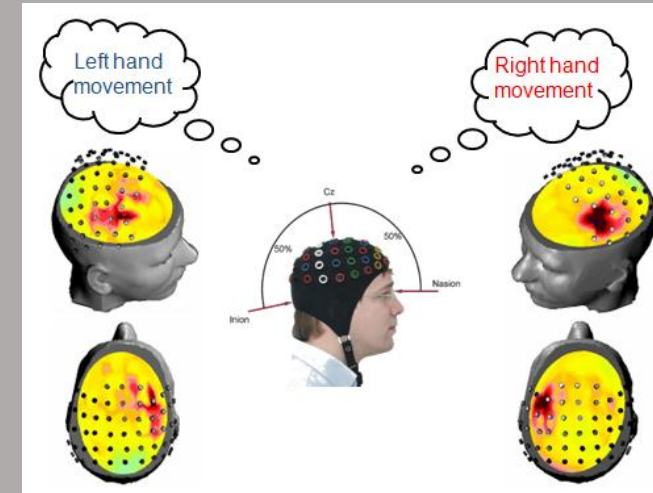


<https://ayazlab.com/research>

Common BCI Paradigms

Motor Imagery (ERD/ERS)

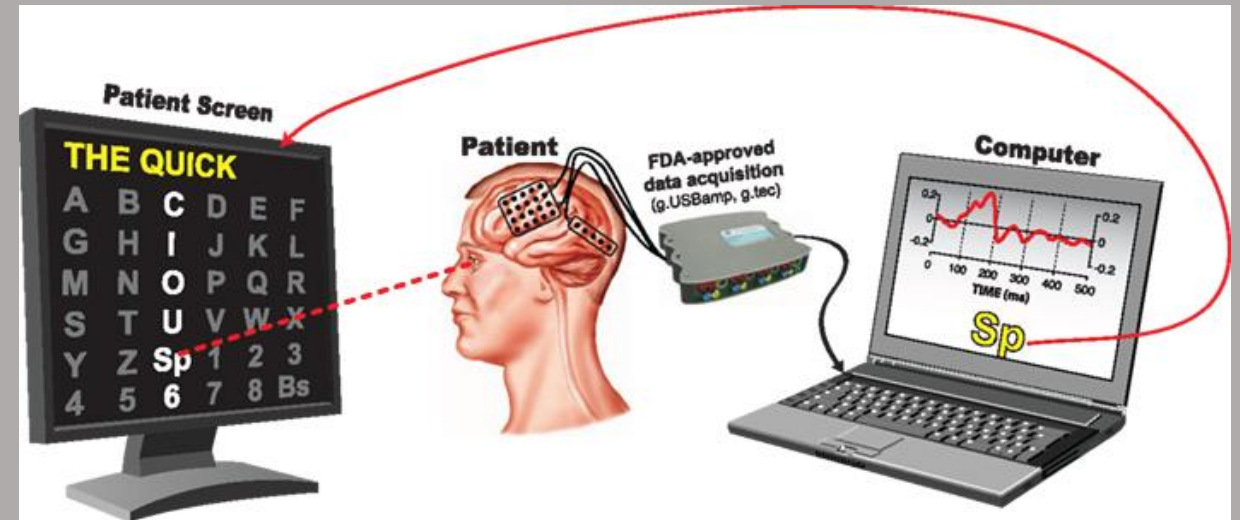
Rolandic mu rhythm (8-12 Hz) and the central beta rhythm (16-24 Hz)



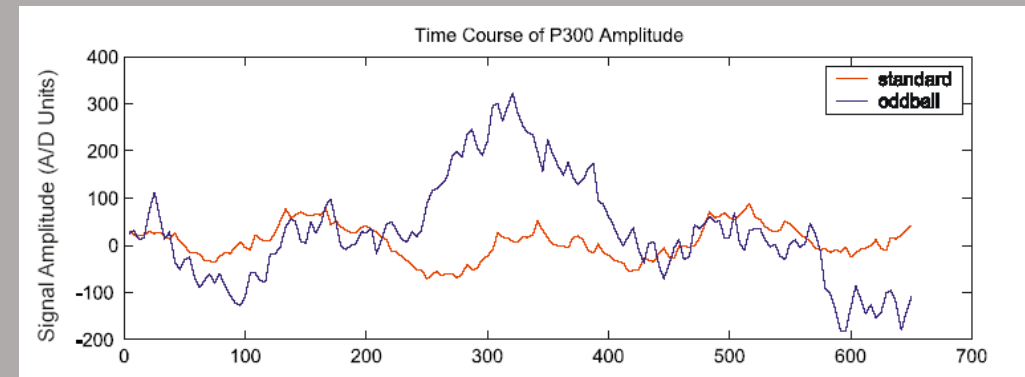
Common BCI Paradigms

P300

Signal associated to the presence of uncommon targets or infrequent stimuli to which a user is paying attention



<https://www.frontiersin.org/articles/10.3389/fnins.2011.00005/full>

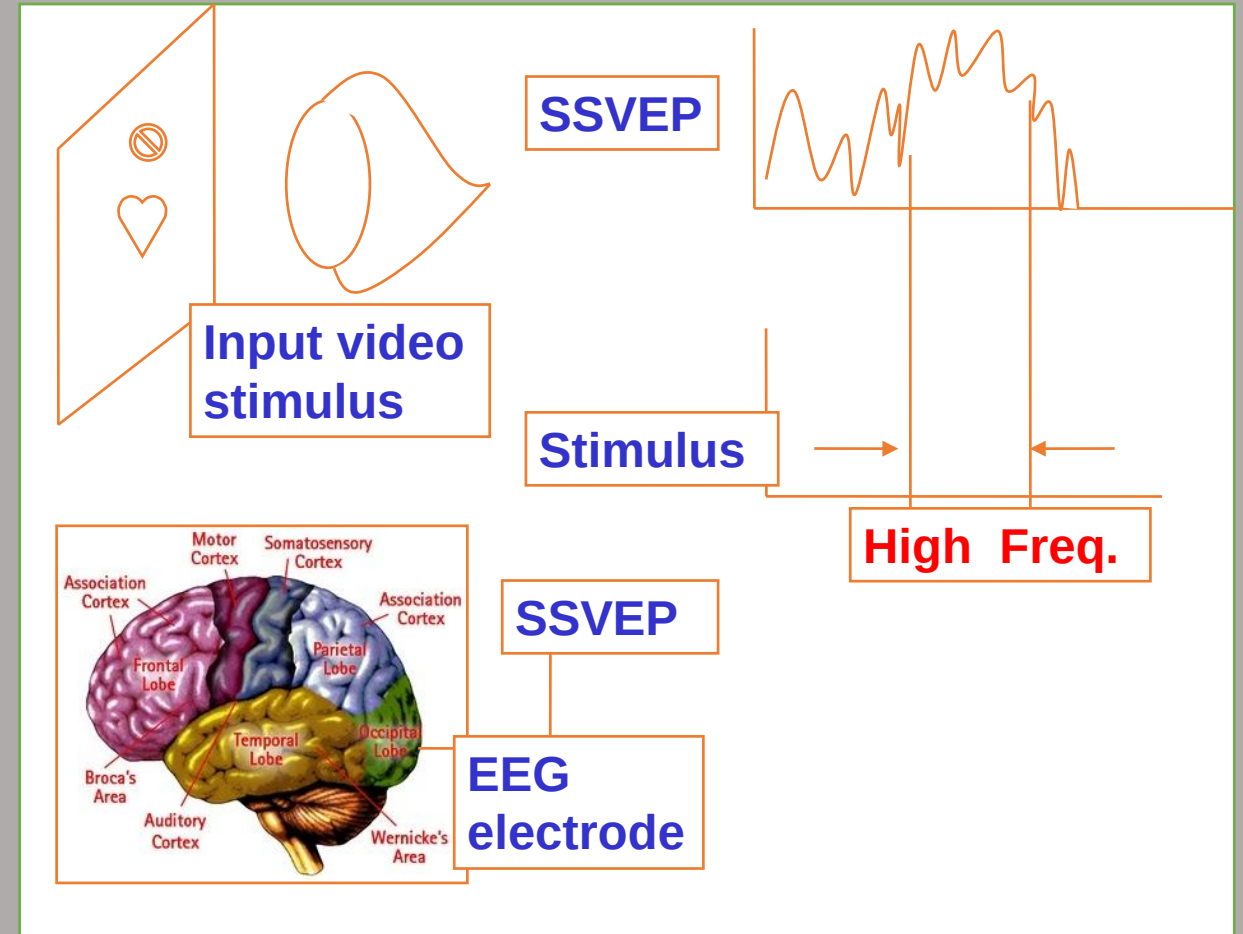


Common BCI Paradigms

Steady-State Visual Evoked Potential

The amplitude of the response is modulated by the frequency of the stimulus.

When participants focus their gaze on a flickering target, the amplitude of the steady-state visually evoked potential (SSVEP) increases at the fundamental frequency of the target and second and third harmonics

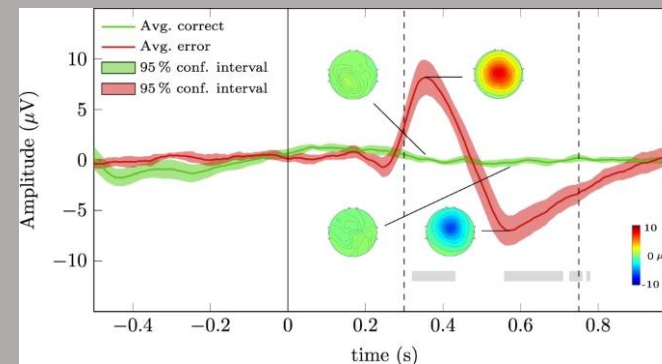
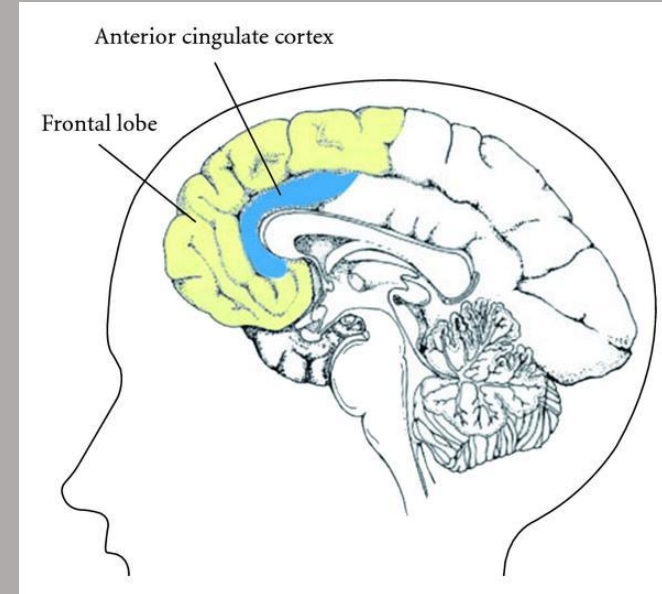


Common BCI Paradigms

Error Related Potential

User's awareness to erroneous response.

Typical occurrences that elicits ErrP:
Choice reaction tasks
Feedback tasks
Observation tasks



BCI Taxonomy

Zander, T.O. & Kothe, C. (2011). "Towards passive brain–computer interfaces: applying brain–computer interface technology to human–machine systems in general." *Journal of Neural Engineering*, 8(2), 025005.

Active BCI: BCIs depend on users deliberately altering their brain or physiological activity via a defined mental strategy to directly control the interface.

Dominant Paradigm – Motor Imagery, Visual Attention

Reactive BCI: BCIs that harness neural responses induced by external stimuli to infer user intentions.

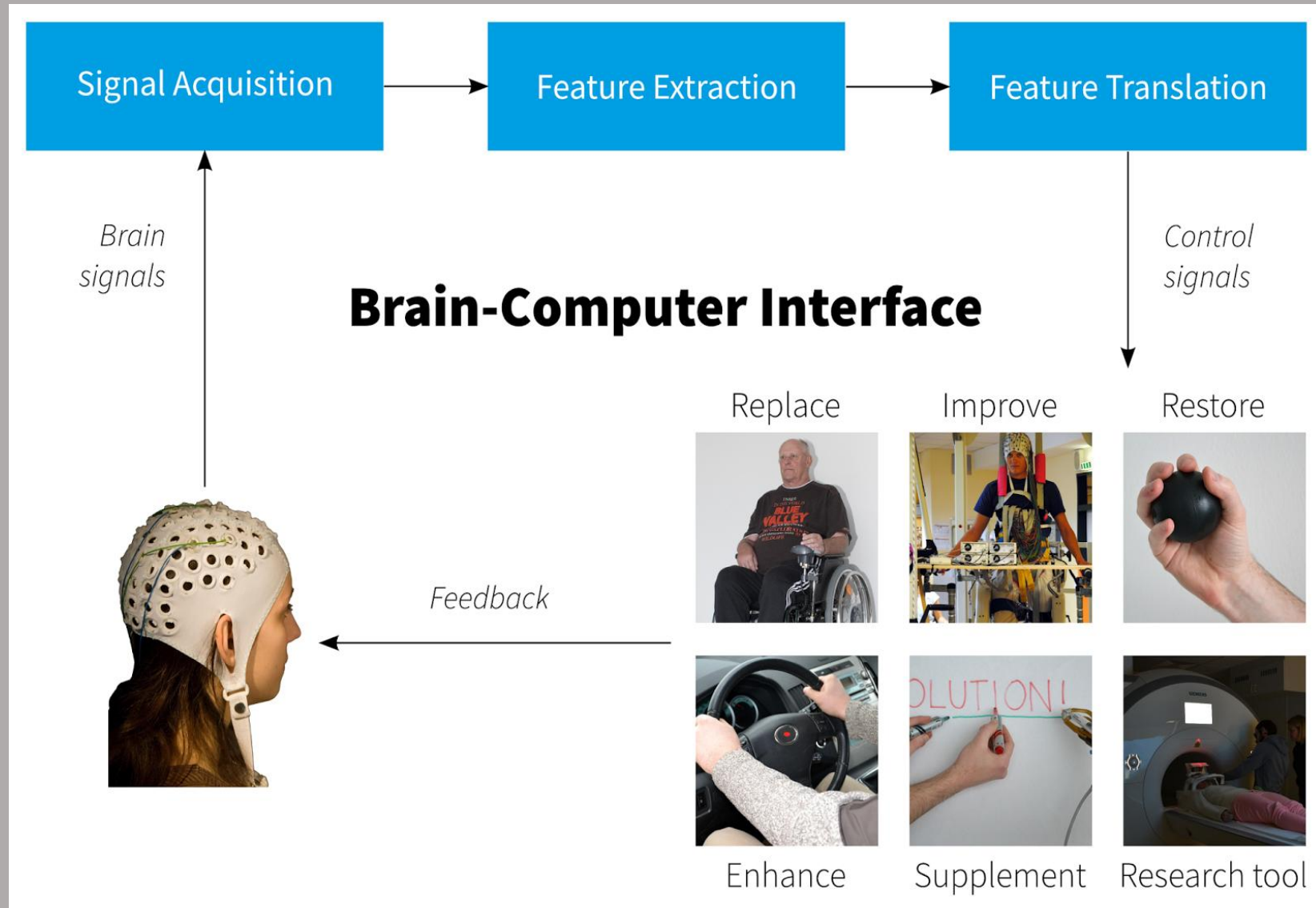
Dominant Paradigm – SSVEP, P300

Passive BCI: BCIs that focus on detecting and quantifying these underlying variables, such as, fatigue, attention, mental workload, affective state.

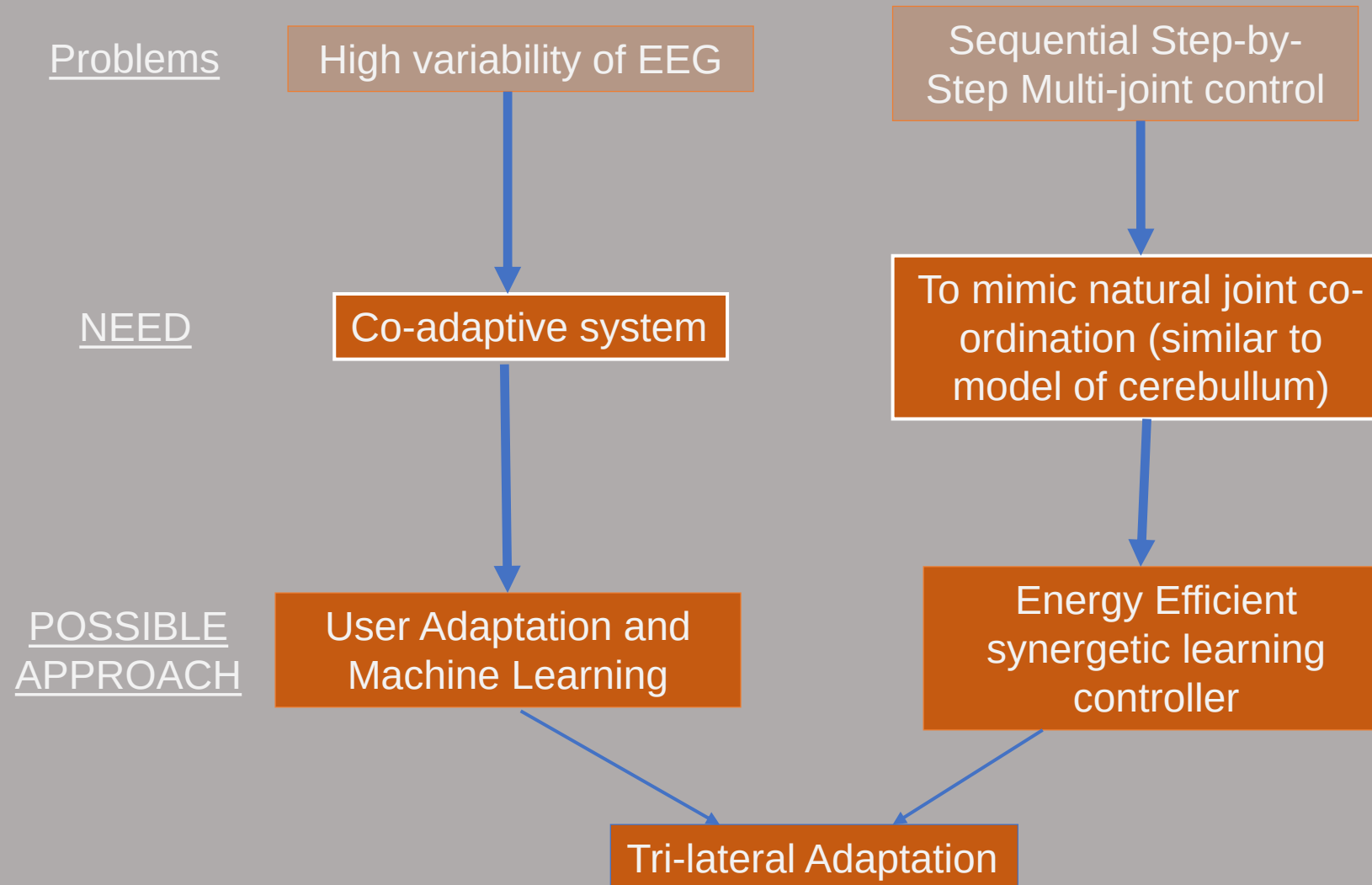
BCI Taxonomy

	Active	Reactive	Passive
Signal Source	Voluntary mental task	Response to external stimuli	Spontaneous, ongoing brain state
User Effort	High, requires training	Low. Dependent on attention	None required.
Example	Wheelchair control	P300/SSVEP Speller	Drowsiness, Workload Detection
Mechanism	Voluntary modulation	Stimulus-locked evoked response	Implicit state monitoring

General Pipeline



BCI for Robotic Arm Control



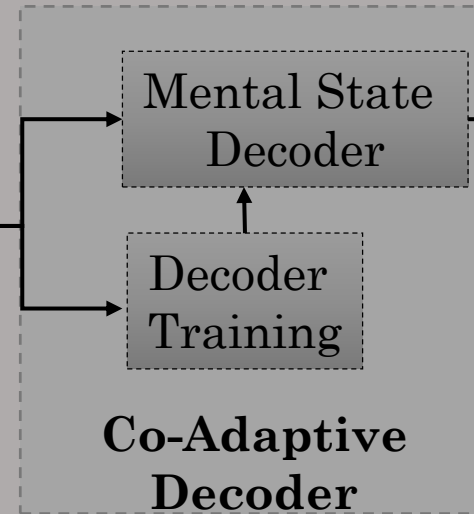
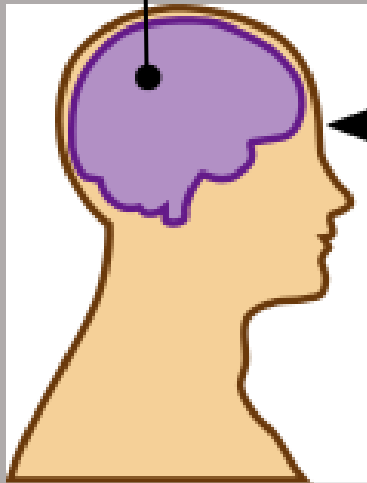
BCI for Robotic Arm Control

Inria

CORTICAL SIDE EEG
ADAPTATION

Pre-processing
And
Feature Extraction

Raw
Motor Imagery
EEG



MOTOR SIDE FOR
ENVIRONMENTAL
ADAPTATION

End-point Motor Intention

Synergetic Motor
Learning Controller

Torque
Command

Proprioceptive
Feedback



Visual Feedback

Up/Down Movement of
end-effector

BCI for Robotic Arm Control

9 participants (3F)

Robotic Arm located in RIKEN
Brain Institute, Japan

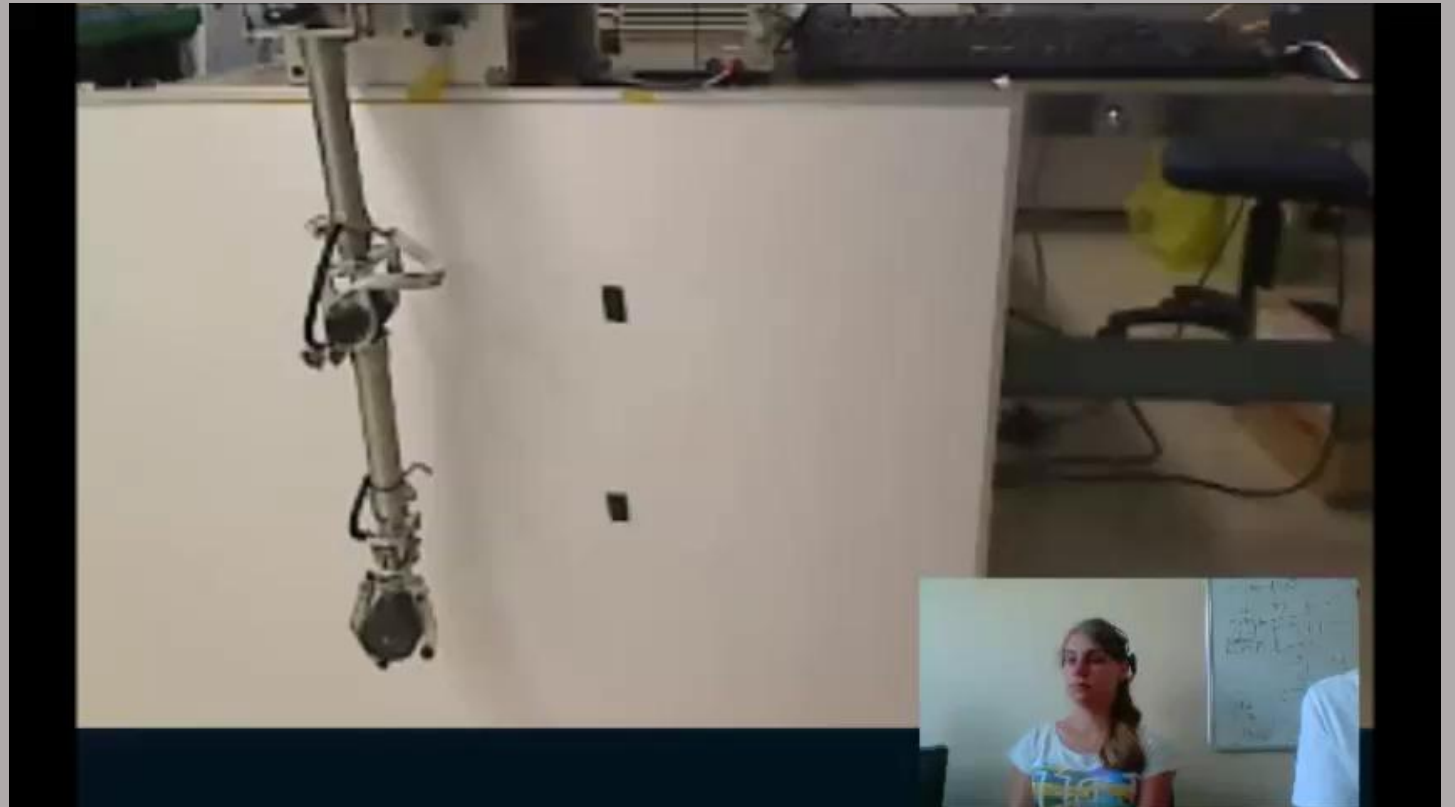
Average transmission time: 8.63sec

BCI Performance

Accuracy: 81.67 ± 6.12

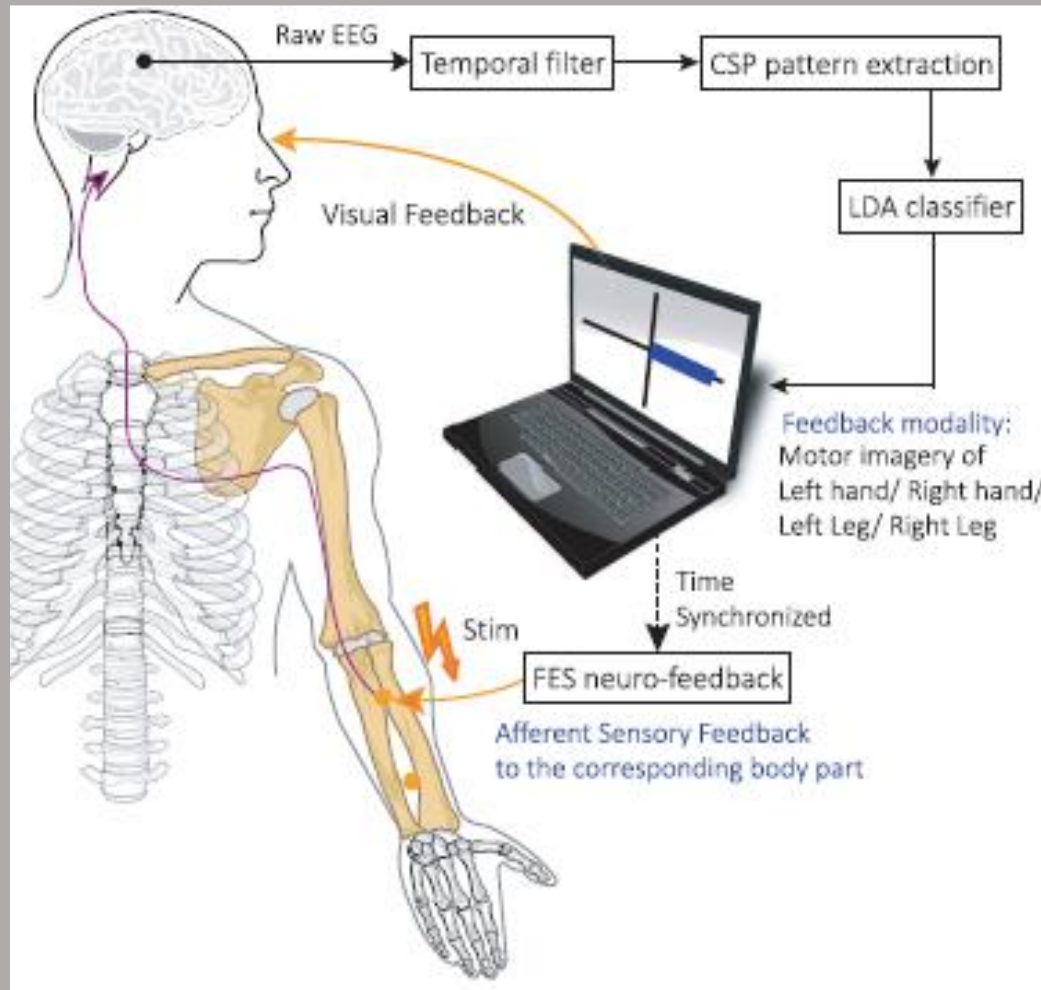
Sensitivity: 83.33 ± 11.18

Specificity: 81 ± 13.65



BCI-FES Control

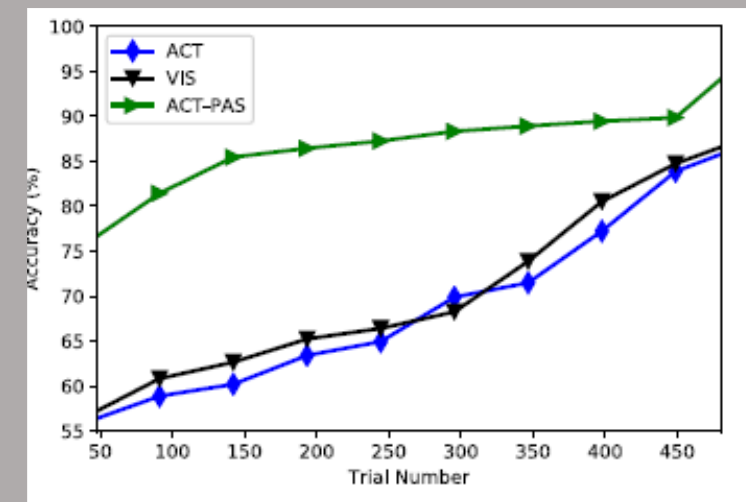
Bhattacharyya et al., *IEEE Trans. Med. Rob. Bionics* 2019



16 participants (3 F)

Mean age: 28 years

Three states – Only FES (ACT), Only VIS (PAS), Combined (ACT-PAS)



BCI-FES for Stroke Recovery

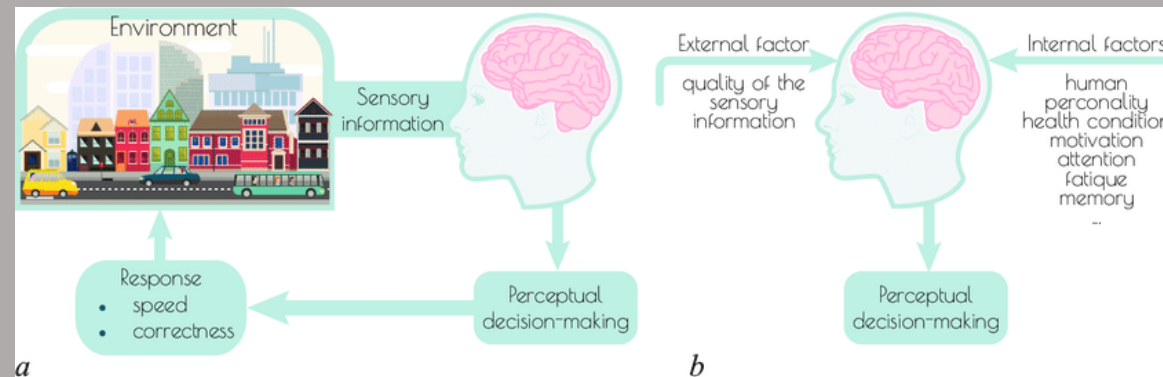
Biasiucci et al., *Nature Comm.* 2018

- **Design:** Randomized controlled trial, 27 chronic stroke survivors with moderate-to-severe upper-limb impairment. 10 sessions of BCI-actuated FES vs. sham FES over 5 weeks. 14 patients in the BCI-FES arm.
- **Task:** Patients attempted hand extension; a motor-imagery BCI decoded the attempt from EEG and, only on correct detection, triggered FES to complete the movement, closing the loop between intention and execution.
- **Result:** BCI-coupled FES produced significant, clinically relevant, lasting motor recovery, more effective than sham FES, with the improvement retained 6–12 months after therapy ended.
- EEG analysis showed a significant increase in functional connectivity between motor areas in the affected hemisphere in the BCI group, and this increase correlated significantly with functional improvement, i.e., the benefit isn't just behavioral, it's traceable to measurable cortical reorganization.

Collaborative BCI for Decision Making

Bhattacharyya et al., *Sci Rep.* 2021

- Billions of decisions are made everyday
- Decision-making → complex cognitive process involving the evaluation of options and the selection of a course of action among several possible alternatives.
 - This process is underpinned by a network of brain regions that interact to assess the value of different outcomes, weigh risks and rewards, and integrate emotional and social factors into the decision-making process.
- Perceptual decision making is the act of choosing one option or course of action from a set of alternatives on the basis of the available sensory evidence.
 - interpreting sensory information to make decisions.



Collaborative BCI for Decision Making

Bhattacharyya et al., *Sci Rep.* 2021

- **Individual Decision accuracy could degrade with:**
 - Limited processing time
 - Quantity and complexity of data
 - Irrelevant information, audiovisual clutters/distractors
 - High-stakes, time-pressured situations
- **Mental state of the operator also affects decision accuracy**
 - Fatigue/alertness
 - Mental workload
 - Attention level

Collaborative BCI for Decision Making

Bhattacharyya et al., *Sci Rep.* 2021

- **Most animals and humans live within groups. Why?**

A group can do things that individuals alone can't:

- **Augmented action capabilities:** Members of a group can join forces to do something that is beyond the strength or endurance of a single individual
- **Increased sensing capabilities:** A group has a much higher probability of finding resources and noticing danger
- **Increased cognition and intelligence:** groups show emergent cognition and intelligent behaviours which are more powerful than those of the individual members.

Collaborative BCI for Decision Making

Bhattacharyya et al., Sci Rep. 2021

- **Group Decision Making**

Power of crowds: groups have augmented wisdom. This is why human decisions are routinely made by committees (where members' knowledge, intelligence, experience and creativity are melded to improve outcomes).

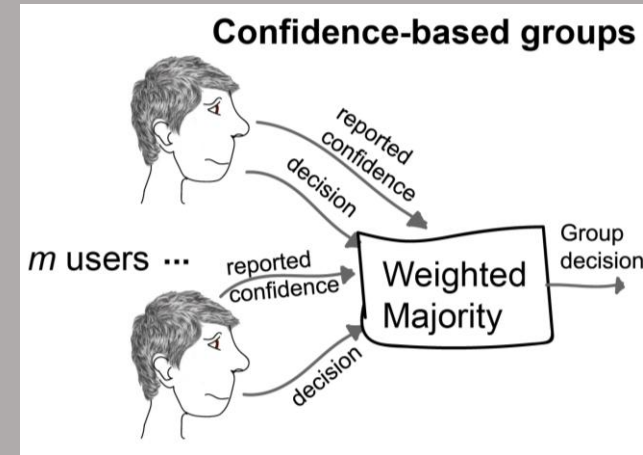
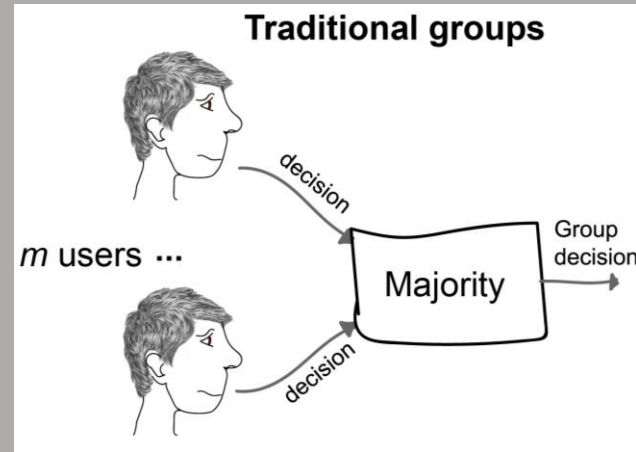
Groups could fail: under- and over-confidence biases, reduced member effort, time constraints, strong leadership, ...



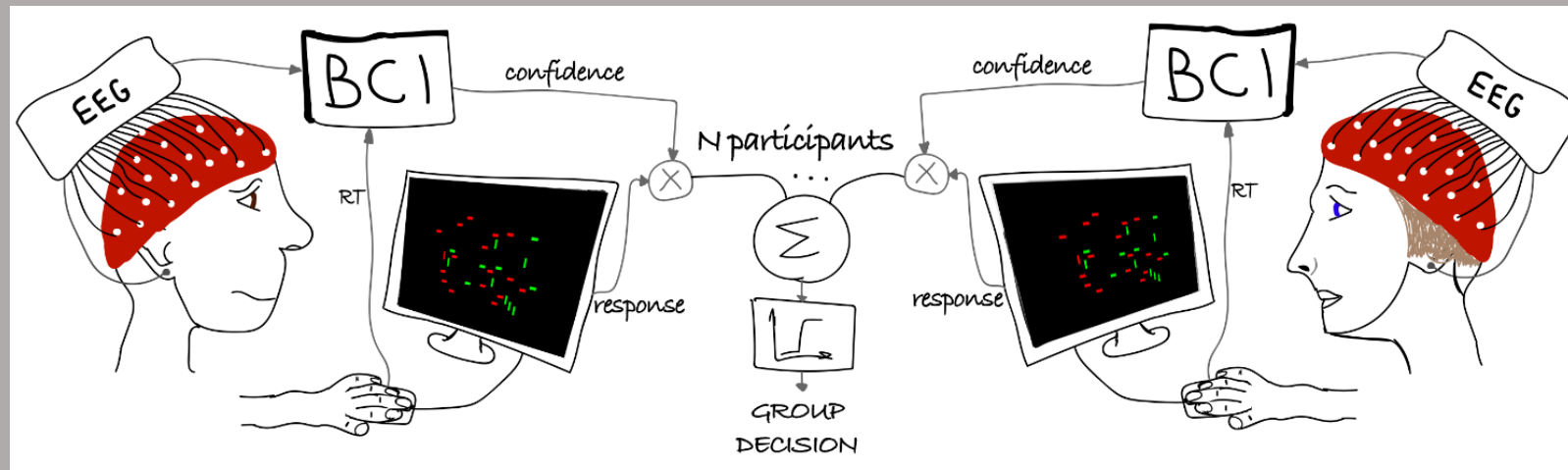
Collaborative BCI for Decision Making

Bhattacharyya et al., *Sci Rep.* 2021

- Group Decision Making Strategies



Our Approach



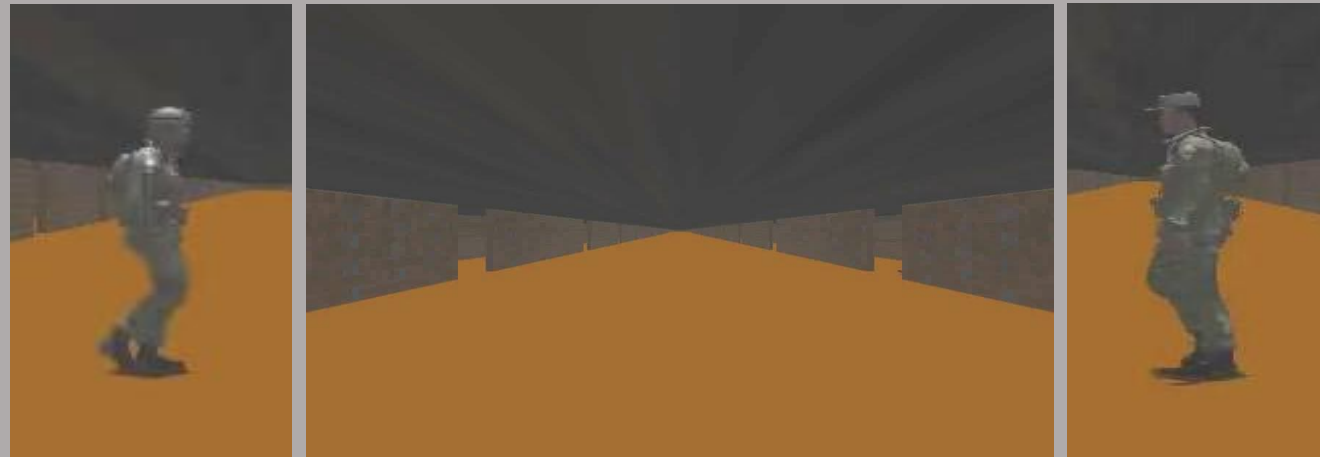
Collaborative BCI for Decision Making

Bhattacharyya et al., *Sci Rep.* 2021

cBCI on Dynamic Environment

➤ Patrol Experiment

- We generated a **dynamic environment** where a soldier is walking along a corridor with multiple doorways present on both sides (**Corridor/Patrol Experiment**)
- **Task:** Decide whether the characters appearing wear a helmet or a cap



Collaborative BCI for Decision Making

cBCI on Dynamic Environment

Bhattacharyya et al., *Sci Rep.* 2021

➤ Outpost Experiment

- User stationed at an outpost. Observes a character moving in.
- Reward and penalties proportional to the correctness of the decision and the time taken by the user to respond
- **Task:** Decide whether the characters appearing wear a helmet or a cap

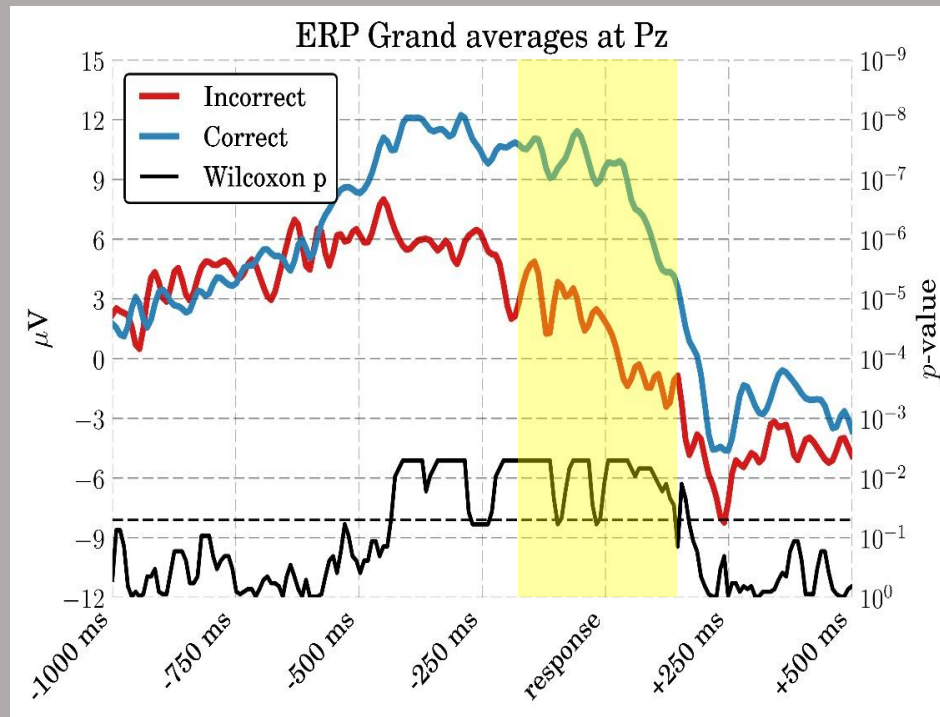


Collaborative BCI for Decision Making

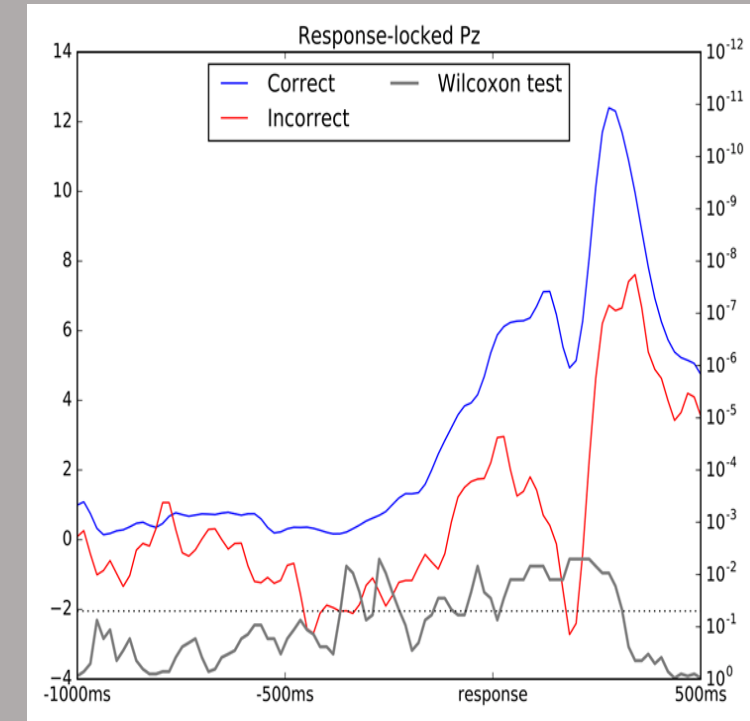
Evidence of Neural Marker

Bhattacharyya et al., *Sci Rep.* 2021

Patrol Experiment



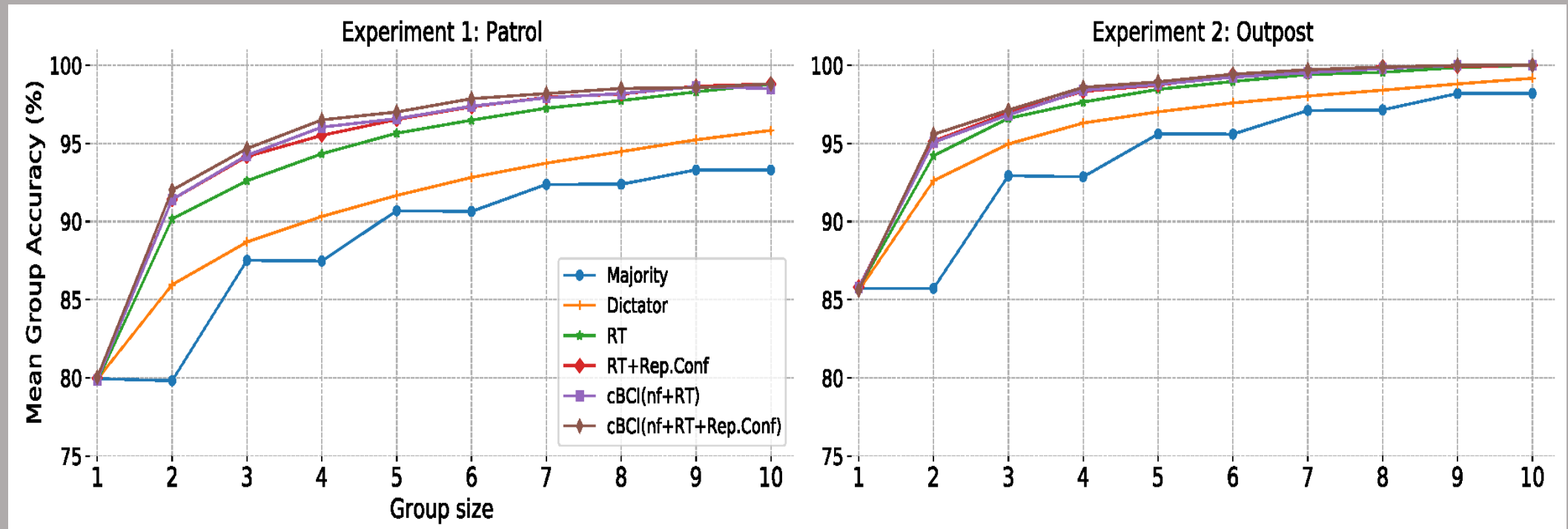
Outpost Experiment



Collaborative BCI for Decision Making

Bhattacharyya et al., *Sci Rep.* 2021

Group Performance



Suggestive Reading

- Biasiucci, A., Leeb, R., Iturrate, I., Perdakis, S., Al-Khodairy, A., Corbet, T., Schnider, A., Schmidlin, T., Zhang, H., Bassolino, M., Viceic, D., Vuadens, P., Guggisberg, A.G., & Millán, J.d.R. (2018). Nature Communications, 9, 2421. doi: 10.1038/s41467-018-04673-z
- Lorach, H., Galvez, A., Spagnolo, V. et al. Walking naturally after spinal cord injury using a brain–spine interface. Nature 618, 126–133 (2023). <https://doi.org/10.1038/s41586-023-06094-5>
- Greenspon, C.M., Valle, G., Shelchkova, N.D. et al. Evoking stable and precise tactile sensations via multi-electrode intracortical microstimulation of the somatosensory cortex. Nat. Biomed. Eng 9, 935–951 (2025). <https://doi.org/10.1038/s41551-024-01299-z>
- Lee, J.Y., Lee, S., Mishra, A. et al. Brain–computer interface control with artificial intelligence copilots. Nat Mach Intell 7, 1510–1523 (2025). <https://doi.org/10.1038/s42256-025-01090-y>
- Ding, Y., Udompanyawit, C., Zhang, Y. et al. EEG-based brain-computer interface enables real-time robotic hand control at individual finger level. Nat Commun 16, 5401 (2025). <https://doi.org/10.1038/s41467-025-61064-x>

Where BCI Still Falls Short

- Signal limitations – inter-subject/inter-session variability. BCI illiteracy (Saha & Baumert (2020) *Frontiers Comp. Neurosci.*, 13)
- Implant safety and biocompatibility (invasive) (Dunlap et al. (2020) *Frontiers Neurorob.*, 14)
-
- Ethics, mental privacy & governance (Yuste et al. (2017) *Nature*, 551)
- Clinical translation gap (IBRO Neuroscience Reports (2025))
- User mental state is treated as noise, not signal → and it's a barrier to real-world adoption (Daniel Klee et al (2026) *J. Neural Eng.*, 23)

Mental States Affects Performance

WHAT WE MEASURE?

Tasks Completed
Accuracy
Response Time

WHAT DRIVES IT ALL?

Attention & Alertness
Mental Fatigue
Cognitive Engagement
Arousal State (Mood)

**Better,
Robust
BCI**

Motivation
Engagement

Mental Fatigue
Attention Level
Cognitive Workload
Arousal State
Engagement

Mental Fatigue – What is it?

Mental fatigue is a latent brain state that develops as a result of **prolonged cognitive effort**. It gradually **impairs attention, decision-making, and performance**, often without conscious awareness.

Reduced Attention

Harder to stay focused

Slower Reactions

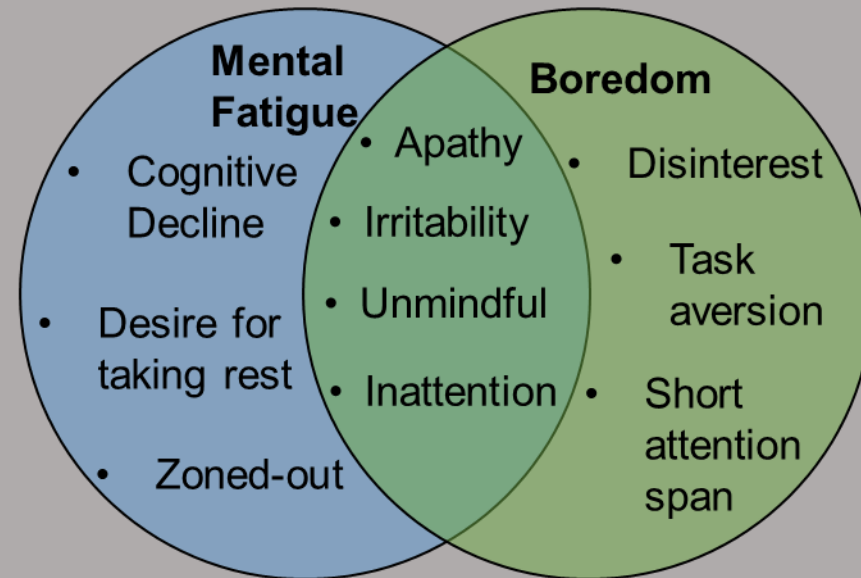
Response time increases, errors go unnoticed

Slower Reactions

Response time increases, errors go unnoticed

Slower Reactions

Response time increases, errors go unnoticed



Why it's hard to solve

- Hard to spot fatigue in real-time → leads to burn-out
- Self-reporting fails
- Everyone is different – varies significantly across individuals
- Poorly defined – overlaps with boredom, low motivation, etc.

OUR RESEARCH QUESTION

Can we **separate mental fatigue from other states**, and then detect its onset objectively, in real time, before the person is even aware of it?

Dissociate, then detect

First, establish what mental fatigue actually is and isn't. Then build a reliable detection method around that

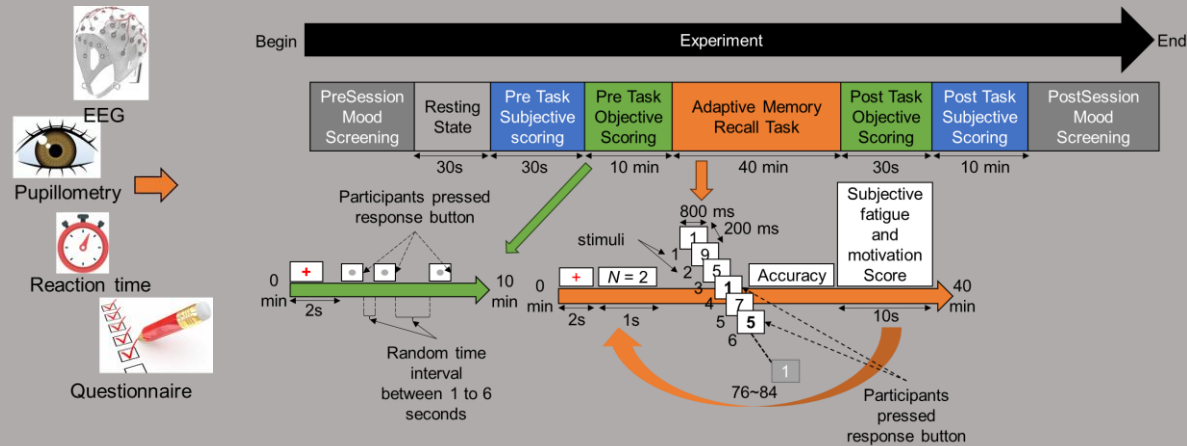
Early Detection, Not Reactive

Detection ahead of awareness.

Time to intervene

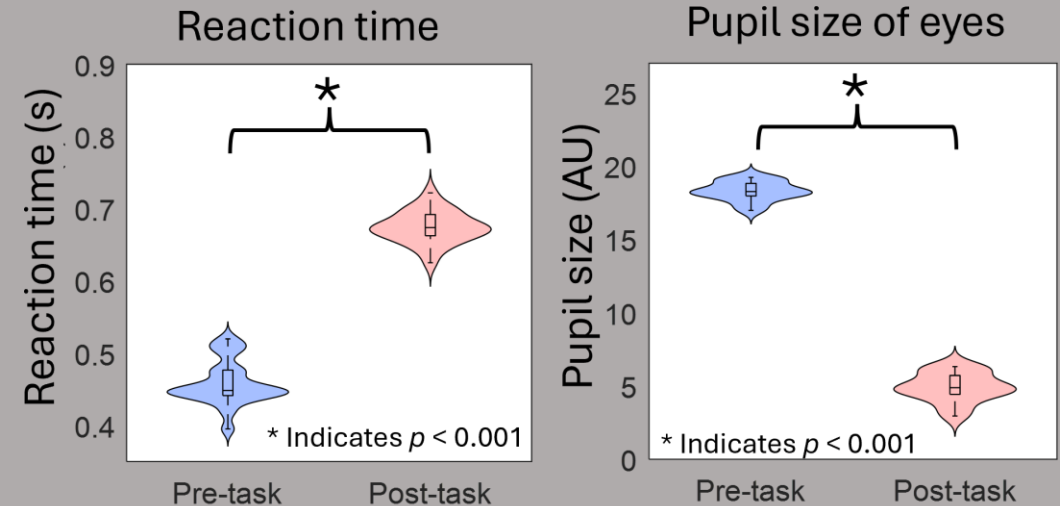
First, we dissociate Mental Fatigue

Samanta et al., *Under Revision*



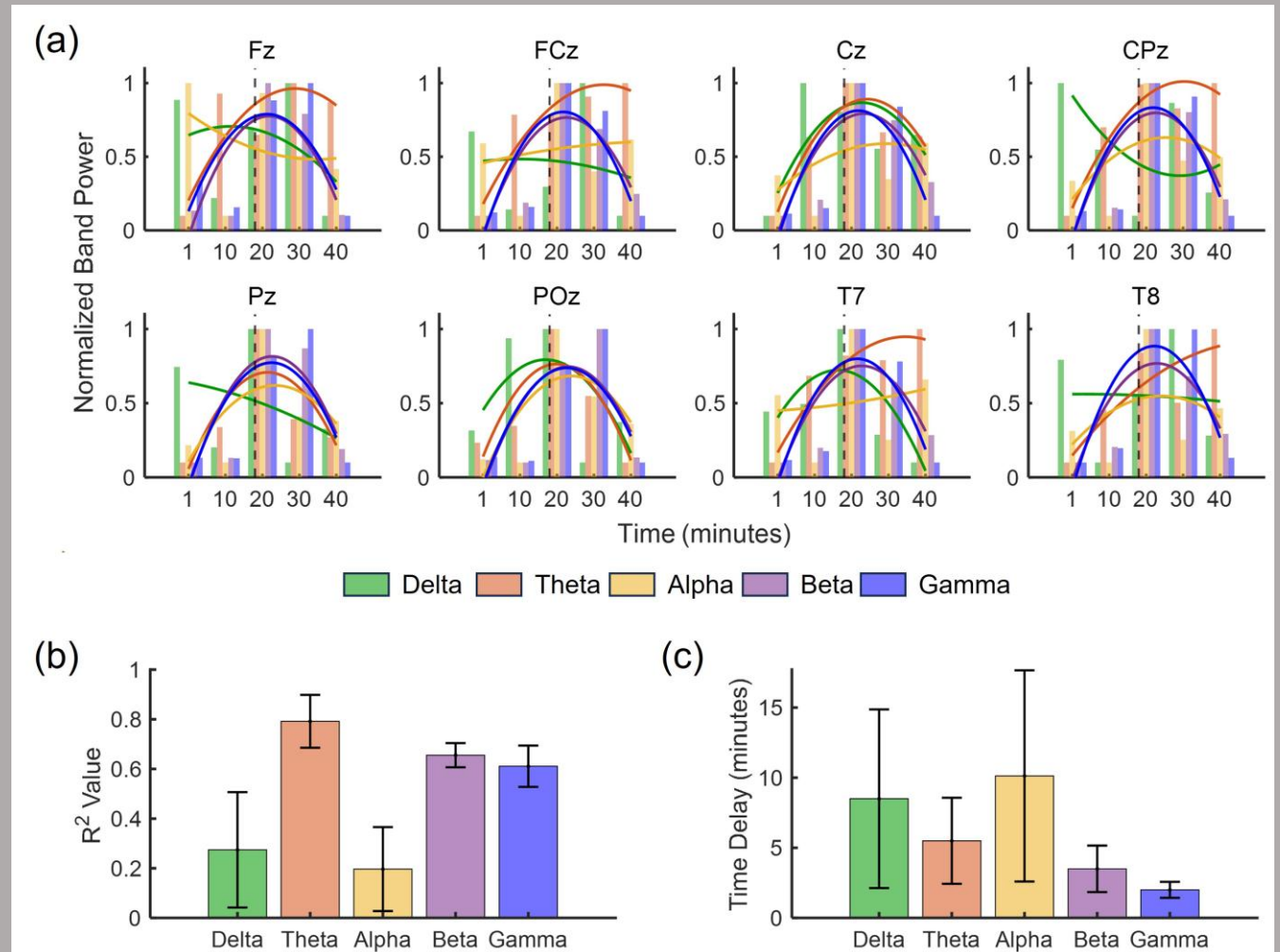
- 22 participants
- **63% participants** reported not at all to moderate level of boredom

- Adaptive Memory Recall Task – **High Cognitive Load**
- Adaptive → Stabilise motivation and boredom



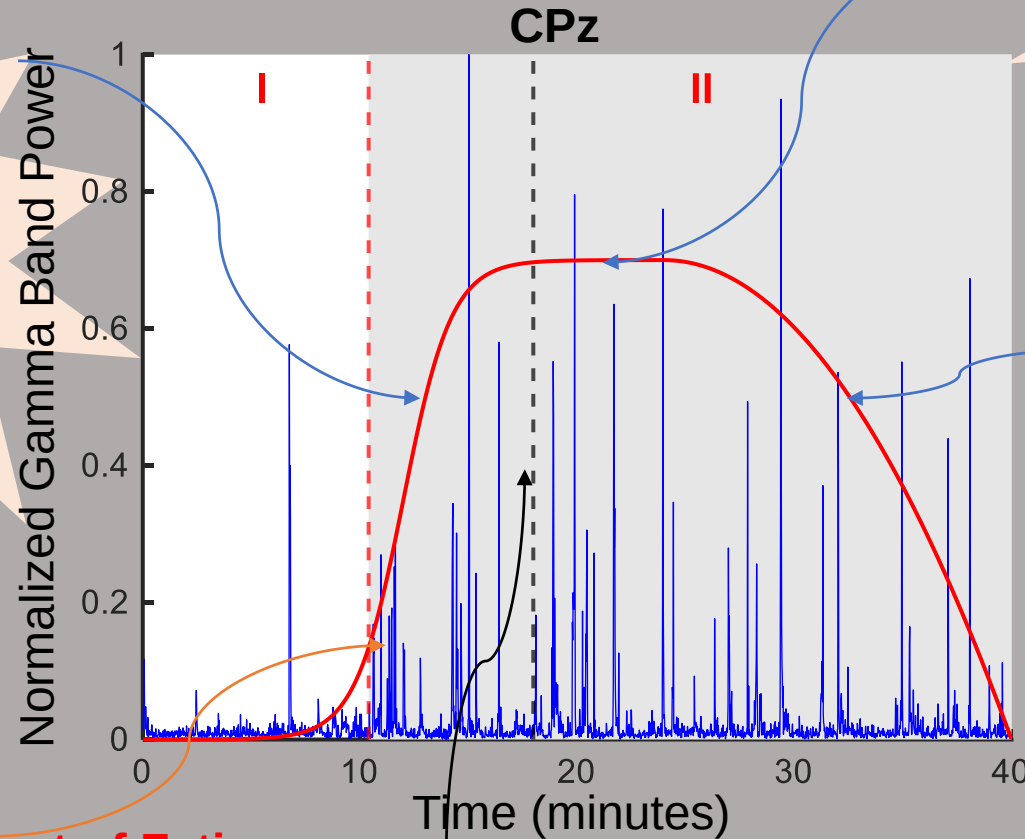
Brain Dynamics

- The peaks are close in temporal proximity to the first instance of the highest self-reported mental fatigue.
- As task time progressed, band power eventually decreased, suggesting a late stage of mental fatigue during which neural resources may become exhausted and task engagement reduced.



Then, we detect Onset of Mental Fatigue

Compensatory mechanism by the brain leading to increased effort



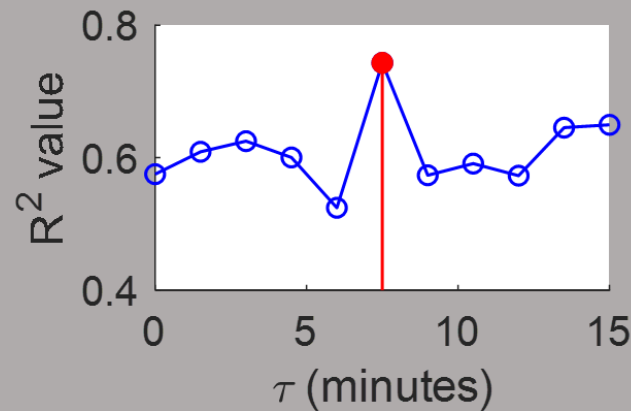
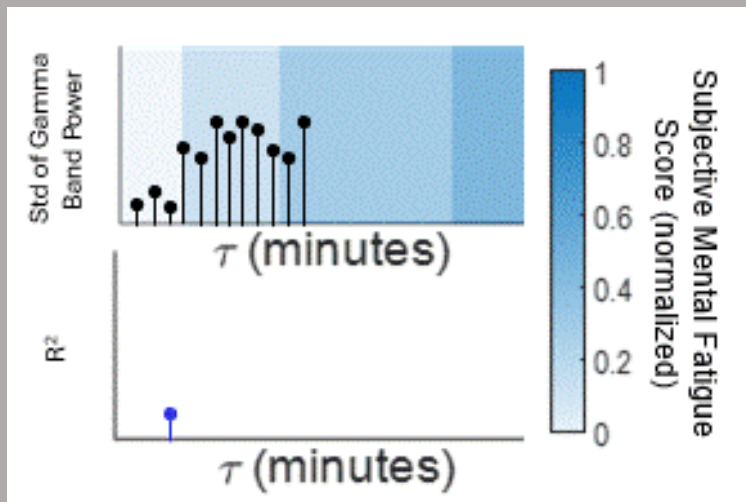
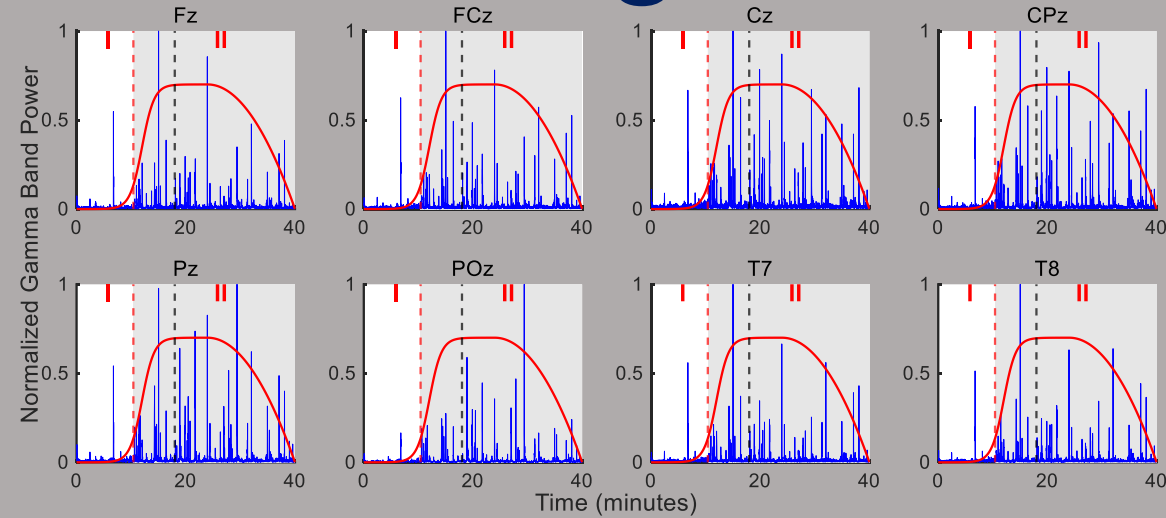
Objective Onset of Fatigue

Highest fatigue reported

Saturation of resources

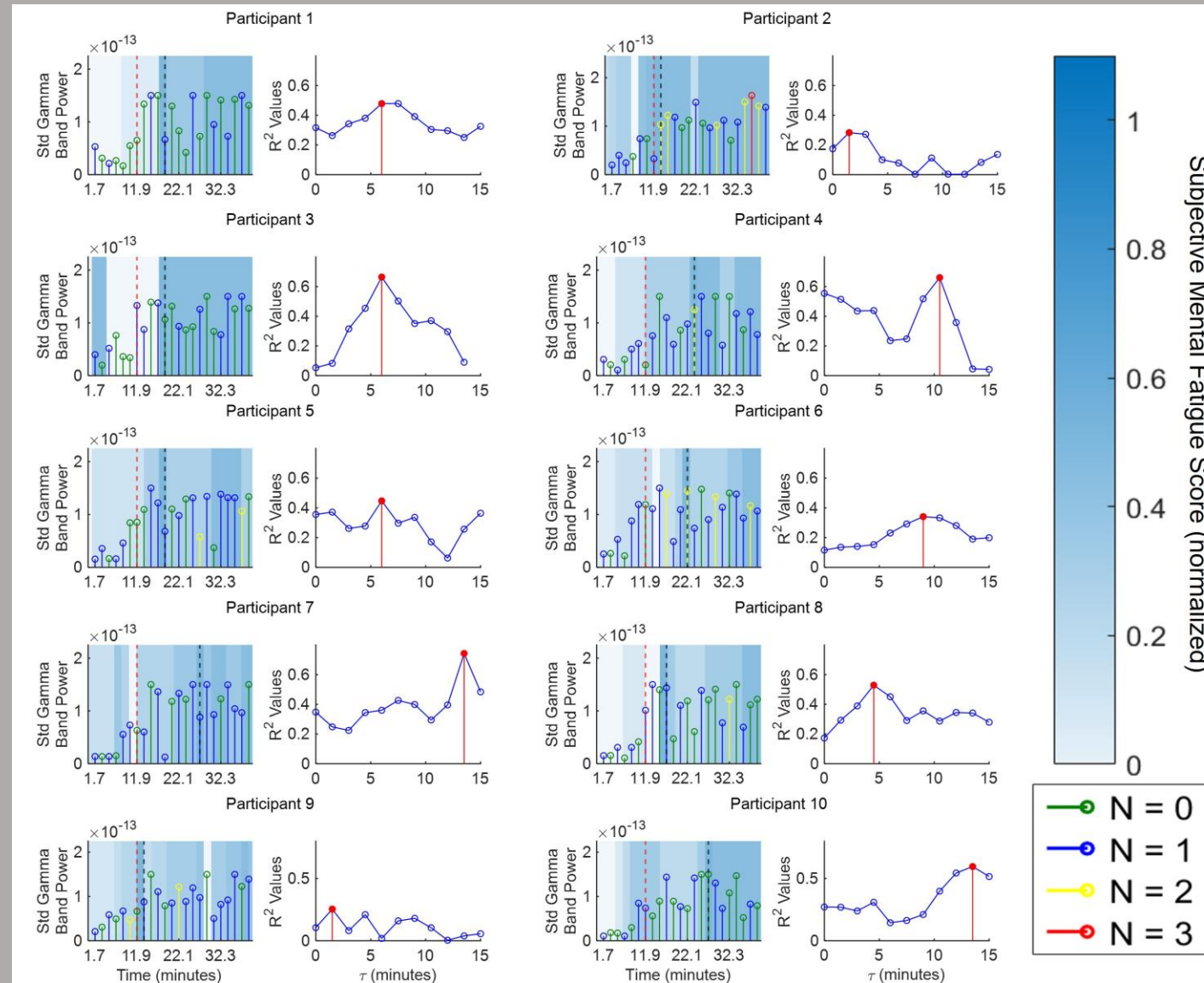
Burnout.
Cognitive
Exhaustion

Consistent across regions

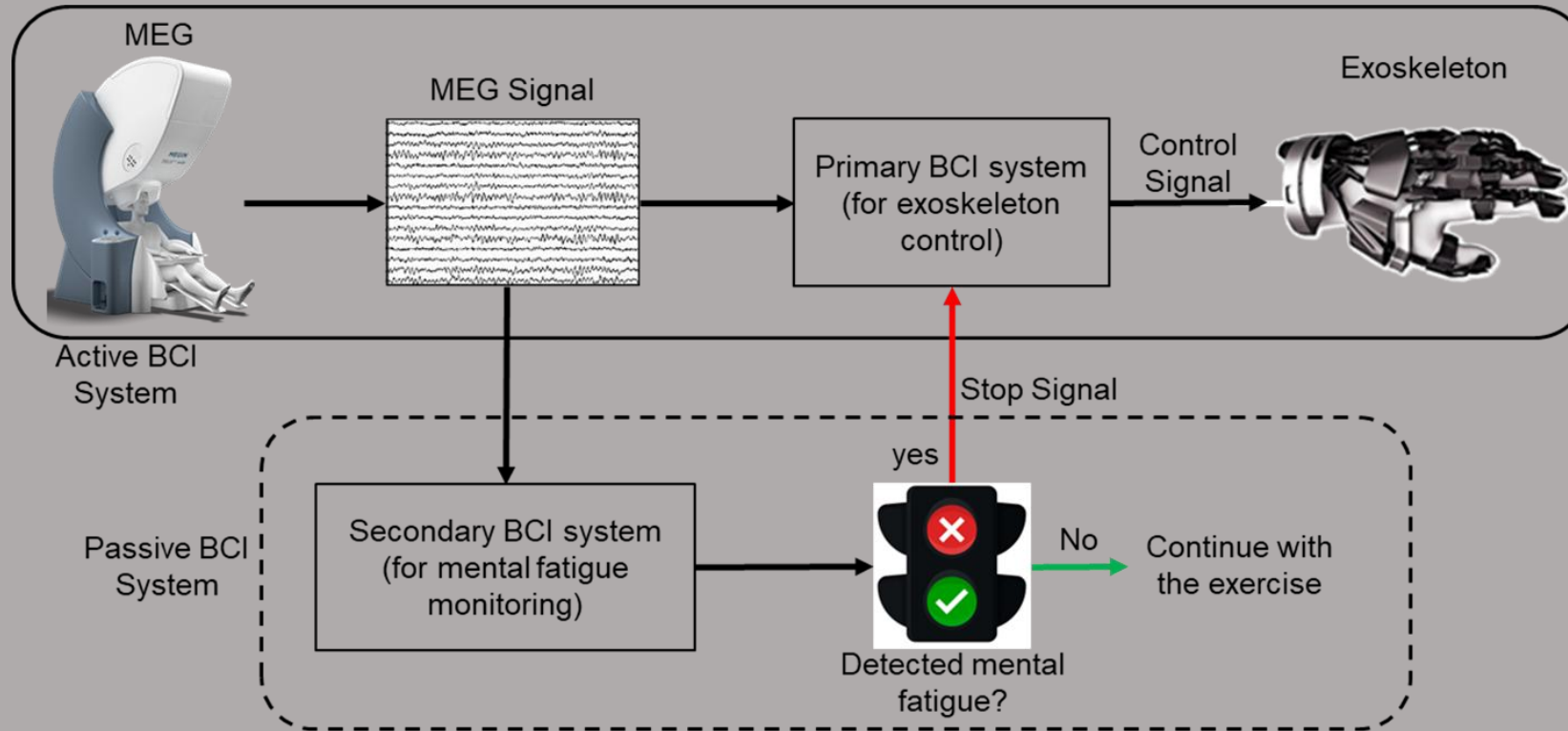


Neurological signatures of the onset of mental fatigue precedes minutes before the person realises (approx. 7.5 minutes).

Individual Time Delay



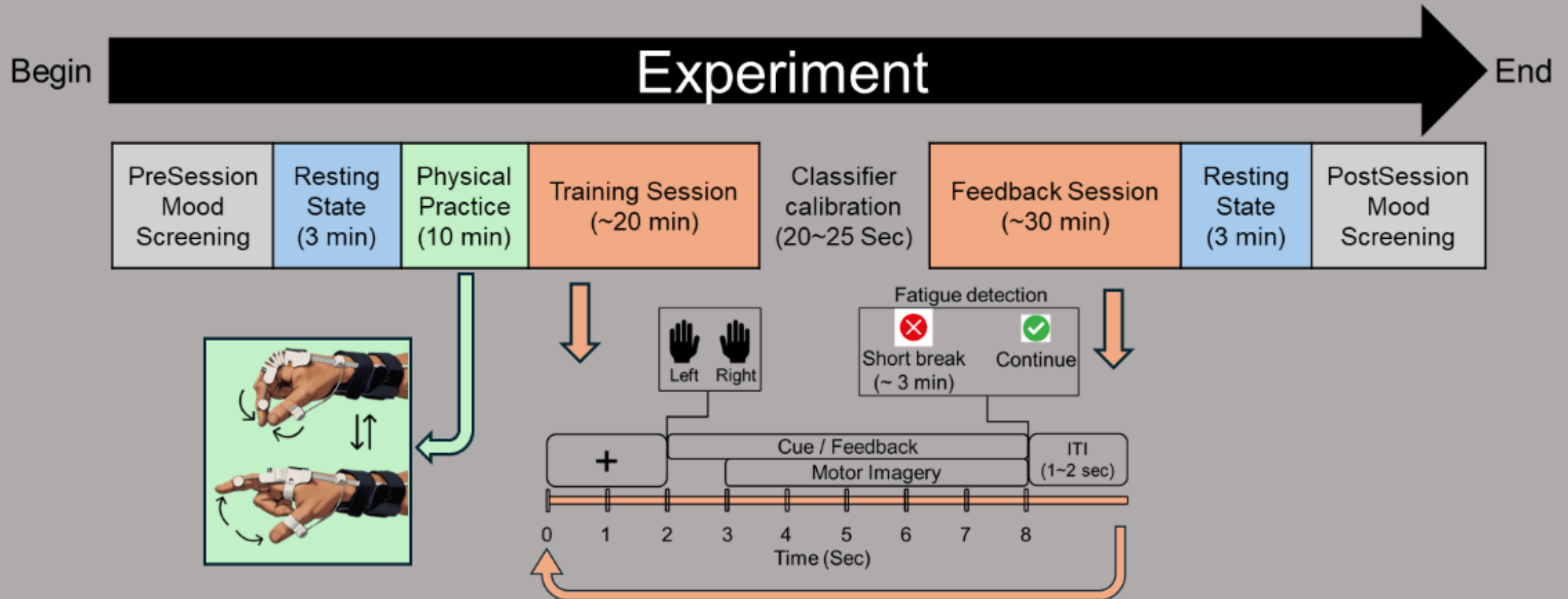
In Motor Imagery Tasks



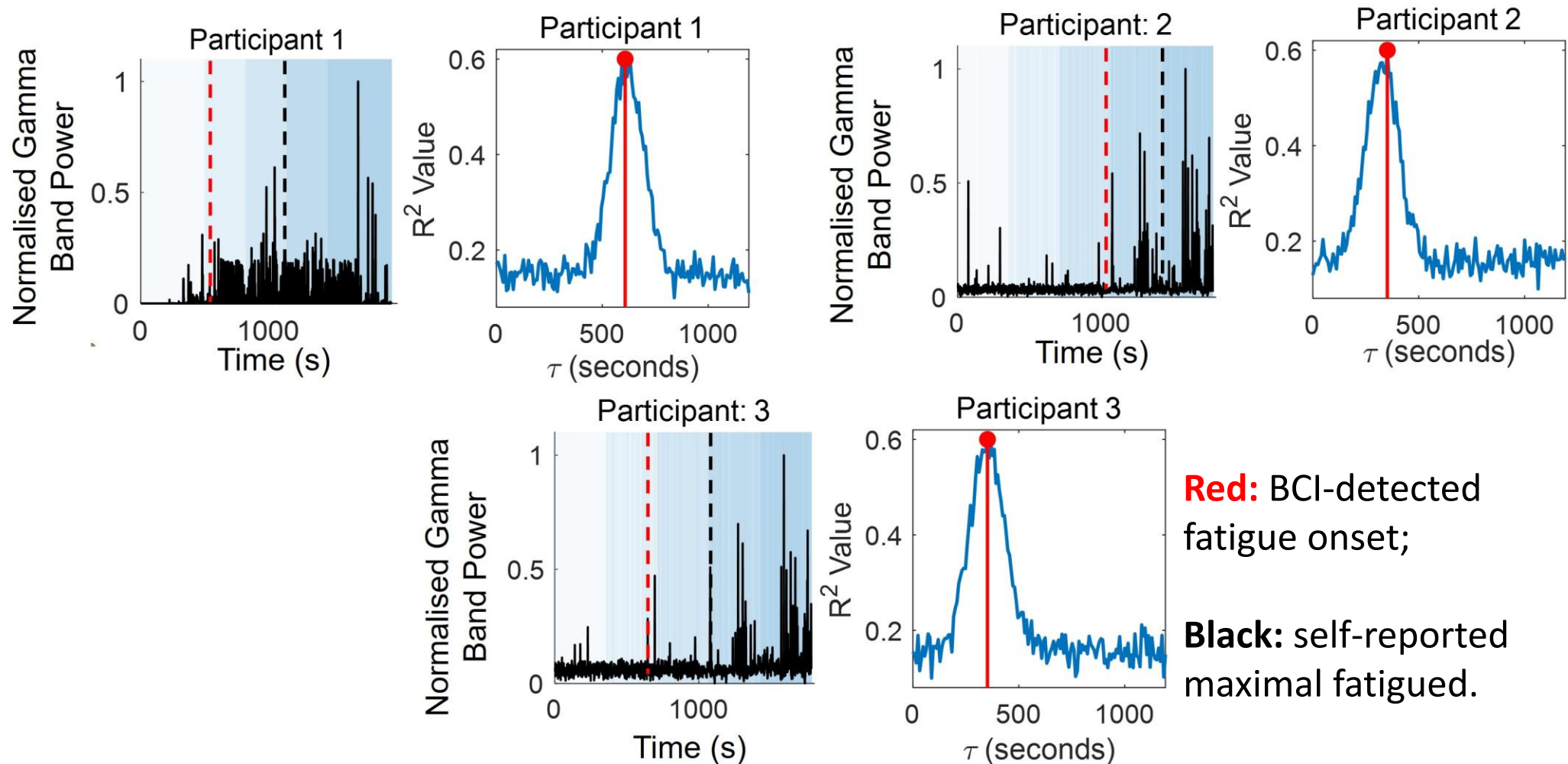
Adaptive modulation of rehab session based on participants' fatigue states

In Motor Imagery (MI) Tasks

Samanta et al., *Under Revision*

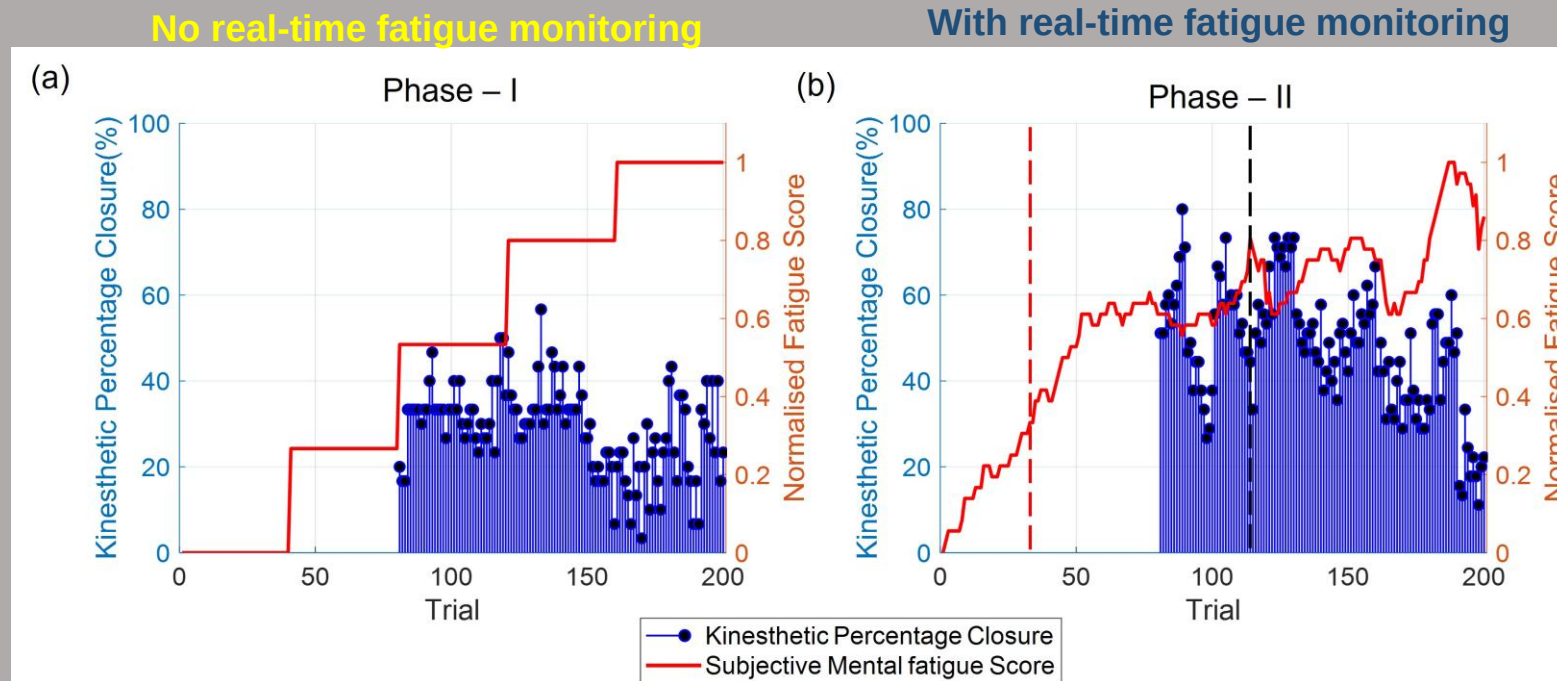


MI-BCI Tracks Subjective Fatigue



MI-BCI Performance

Each movement of the Exoskeleton arm moves in 5 stages from open to close which has been recorded and graphed as kinesthetic percentage closure (KPC)

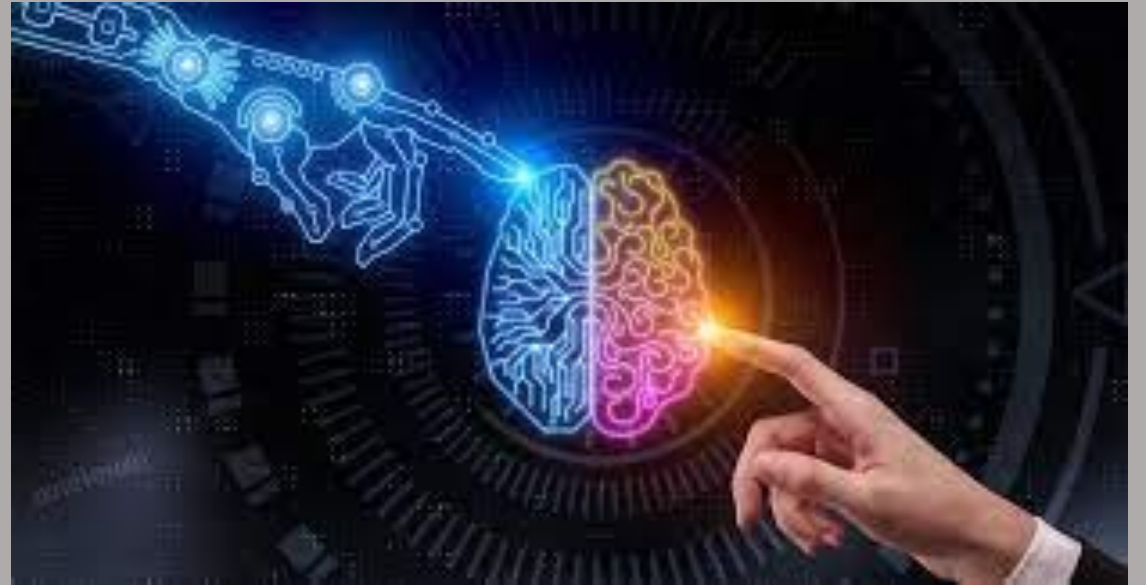


- Addition of Fatigue Monitoring System → Improvement in performance
- Dual-BCI approach helps participants to recover from mental exhaustion.

Summary

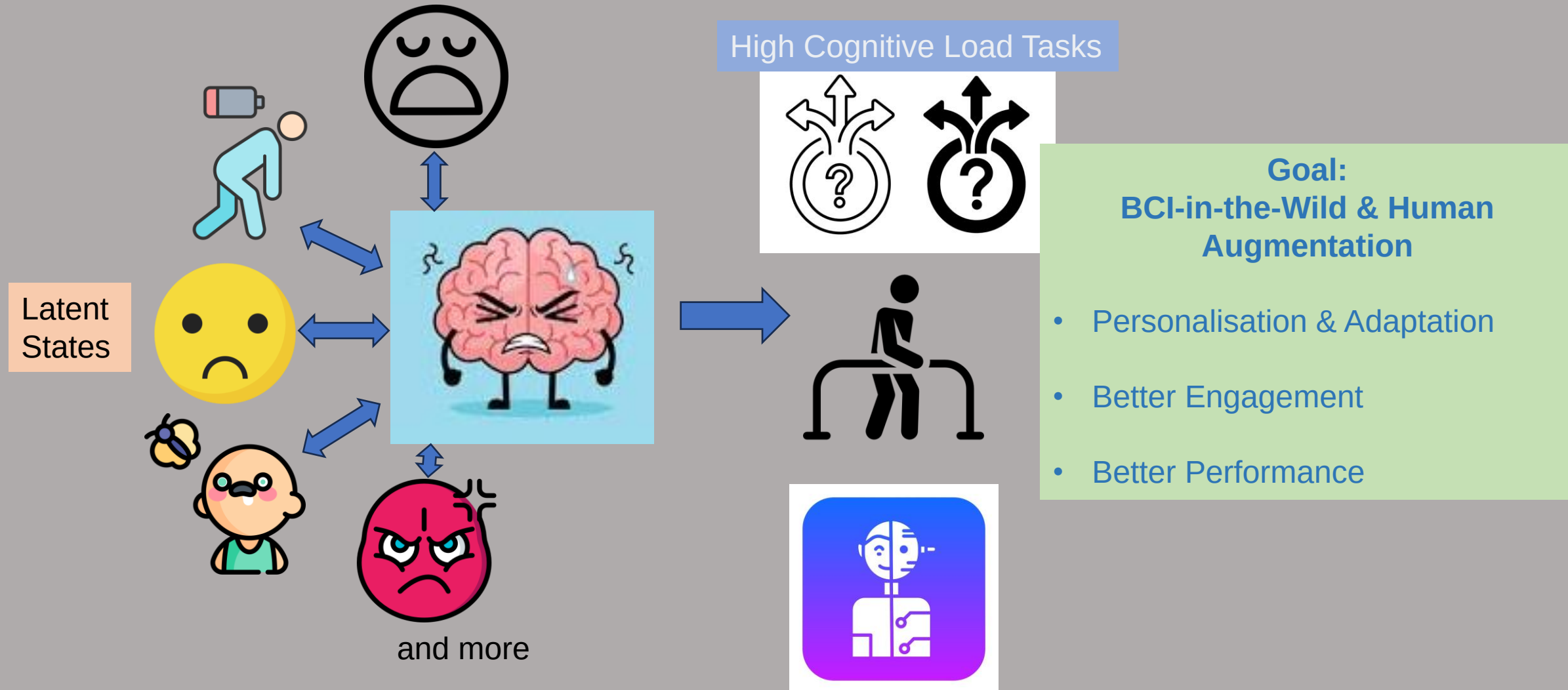
- Neurological signatures of the onset of mental fatigue **significantly precedes** minutes before the person realises (approx. 7.5 minutes in the Memory Recall Task).
- Signatures are **task-**, **age-** and **condition-**agnostic
- Personalised BCI can **help regulate mental exhaustion level** during a high cognitive load task, thereby improving performance and engagement.
- Evidence of neuroplasticity: **Functional connectivity increased progressively** over several weeks for both participants, **indicating positive neurological adaptation alongside therapy**. (Further Investigation Ongoing)

Thanks to the Team



s.bhattacharyya@ulster.ac.uk

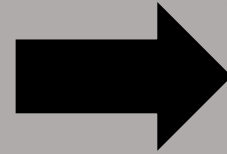
Next Steps - Discovery



Next Steps – Translation & Potential Applications



High Resolution M/EEG



Commercial, Scalable EEG



Eye tracking systems

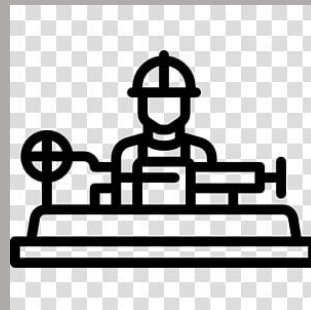


HRV

Non-clinical applications, we foresee



Know when driver needs a break before they do.



Catch the error before it happens



An extra set of eyes on your pilot's cognitive state without adding to their workload.



- *Smarter scheduling of high-stakes tasks*
- *Workload distribution*

Some questions to think about

- We think the signatures are task-, age-, and condition-agnostic. What would it take to convince you they generalize to a completely different domain, say a purely physical fatigue task, or an emotional-regulation task?
- Our personalized BCI can suggest a driver or pilot when they need a break. At what point does that shift from 'helpful assistant' to 'surveillance of your mental state by your employer'? Where's the line?
- Where do you think general-use BCI will actually be in five years, closer to everyday wearables monitoring our mental state, or still mostly confined to the lab and clinic?

Thank you

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Open to Collaboration!!